

Institute of Actuaries of India

Subject CT8 – Financial Economics

April 2016 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) Market over reacts to a news and corrects itself over couple of trading sessions. – Contradicts all forms of efficiency because as per the EMH it is expected that the information is reflected in the stock price immediately. Over reaction or steady correction cannot happen in efficient markets. [1]
- ii) An investor earns returns that are significantly higher than those predicted by a log-normal model fitted for a particular index using technical analysis. – There are two possibilities here: Either the log normal model is not applicable to the index in question in which case nothing can be commented on market efficiency OR the markets may be inefficient if the modeling is correct as it is not possible to consistently earn excess over the risk adjusted returns. [2]
- iii) Equity yields are correlated with the yields implied in the last three trading sessions and show an element of mean revision over a 4-5 year timeframe. – When data for 3 sessions is considered, it contradicts weak form of market efficiency. As per the weak form knowledge of price history should not be of any use if last price is known. However the auto-correlation does add information about the behaviour of share price. The mean revision can also be due to change in investors' risk return preferences and no comments can be made for the 4-5 year period.

[2]

[5 Marks]**Solution 2:**

- i) Let the continuously compounded rate of return per annum realized between time zero and T be λ .

It follows that $S_T = S_0 \exp(\lambda T)$ which means $\lambda = \frac{1}{T} \left(\ln \frac{S_T}{S_0} \right)$

Assumption: S_t follows Geometric Brownian motion with drift μ and volatility σ

This implies $S_t = S_0 e^{\mu t - \frac{1}{2}\sigma^2 t + \sigma W_t}$

Where $W_t \sim N(0, \sigma^2 t)$ i.e. mean = 0 and variance $\sigma^2 t$

Implies $\frac{S_t}{S_0} = e^{\mu t - \frac{1}{2}\sigma^2 t + \sigma W_t}$

Or $\ln \frac{S_t}{S_0} = \mu t - \frac{1}{2}\sigma^2 t + \sigma W_t$

Or $\ln \frac{S_t}{S_0} \sim N\left(\mu t - \frac{1}{2}\sigma^2 t, \sigma^2 t\right)$

This implies rate of return λ follows $N\left(\left(\mu - \frac{1}{2}\sigma^2\right), \frac{\sigma^2}{T}\right)$

This implies Expected value of rate of return = 12.5% and standard deviation 21.21%

[4]

ii) S_t follows Geometric Brownian motion with drift μ and volatility σ

This implies

$$S_t = S_0 e^{\mu t - \frac{1}{2}\sigma^2 t + \sigma W_t}$$

Where $W_t \sim N(0, \sigma^2 t)$

From the above equation of stock price S_t , we can express $S_t = f(W_t, t)$

Using Itô's Lemma, we have

$$\begin{aligned} dS_t &= \frac{\partial f}{\partial W_t} dW_t + \frac{1}{2} \frac{\partial^2 f}{\partial^2 W} (dW_t)^2 + \frac{\partial f}{\partial t} dt \\ &= S_0 e^{\mu t - \frac{1}{2}\sigma^2 t + \sigma W_t} \sigma dW_t + \left(\frac{1}{2}\right) S_0 e^{\mu t - \frac{1}{2}\sigma^2 t + \sigma W_t} \sigma^2 dt + S_0 e^{\mu t - \frac{1}{2}\sigma^2 t + \sigma W_t} (\mu - \frac{1}{2}\sigma^2) dt \\ &= S_t \sigma dW_t + \left(\frac{1}{2}\right) S_t \sigma^2 dt + S_t (\mu - \frac{1}{2}\sigma^2) dt \\ &= S_t \sigma dW_t + S_t \mu dt \end{aligned} \quad [3]$$

iii) $F = f(S)$, from Itô's Lemma we have $d(F) = \frac{\partial f}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial^2 S_t} (dS_t)^2 + \frac{\partial f}{\partial t} dt$

$$\begin{aligned} &= \frac{\partial f}{\partial S_t} (S_t \sigma dW_t + S_t \mu dt) + \frac{1}{2} \frac{\partial^2 f}{\partial^2 S_t} (S_t)^2 \sigma^2 dt + \frac{\partial f}{\partial t} dt \\ &= \left(\frac{\partial f}{\partial S_t} S_t \mu + \frac{1}{2} \frac{\partial^2 f}{\partial^2 S_t} (S_t)^2 \sigma^2 + \frac{\partial f}{\partial t} \right) dt + \frac{\partial f}{\partial S_t} S_t \sigma dW_t \end{aligned}$$

where $\frac{\partial f}{\partial S_t} S_t \mu + \frac{1}{2} \frac{\partial^2 f}{\partial^2 S_t} (S_t)^2 \sigma^2 + \frac{\partial f}{\partial t}$ is the drift and $\frac{\partial f}{\partial S_t} \sigma$ is the volatility of F .

Substituting $\sigma = \sigma_S$ from the question, we have $\sigma_F = \frac{\partial f}{\partial S_t} \sigma_S$

When the growth rate μ on the stock S increases by $\pi \sigma_S$, the drift of F increases by

$$\frac{\partial f}{\partial S_t} S_t \pi \sigma_S = \pi \sigma_F F$$

[5]

[12 Marks]

Solution 3:

	Share Price (Rs.)	Returns	Volatility
Share A	100	10%	10%
Share B	200	12%	25%

An investor wants to invest Rs. 10000 for one year

i) The proportion that needs to be invested in A for minimum variance condition is

$$x_A = \frac{V_A - C_{AB}}{V_A + V_B - 2C_{AB}} = 0.756$$

This implies Rs. 7560 in share A and balance (Rs. 2440) in share B

Since infinitely divisible holdings are not allowed, the investor will have to invest Rs. 7600 in share A and Rs. 2400 in share B.

The expected Return in this portfolio is $x_A R_A + x_B R_B = 10.48\%$

[3]

ii) When the correlation coefficient is 0.6, the minimum variance condition will reveal negative holding of one of the stock.

Since short selling is not permitted in the market, the minimum variance will be achieved when portfolio has only share A. [2]

- iii) For the Modern Portfolio theory to be applicable, the investor should be non-satiated and risk averse. [1]

[6 Marks]

Solution 4:

- i) Would a prior investment decision that resulted in a loss stop you from making a similar decision, even if the new investment appears to be the best alternative?

- Regret Aversion

[1]

- ii) How frequently do you review your investment portfolio?

- Status Quo Bias

[1]

- iii) A popular analyst has commented that markets are expected to provide returns in excess of 15% in the coming year. Will you invest in equities with a hope to earn at least 15% for next year?

- Anchoring

[1]

- iv) Would you prefer to manage your portfolio yourself or rely on views of expert analysts?

- Over confidence.

[1]

[4 Marks]

Solution 5:

- i) Let π be the value of the derivative such that $\pi(S, t) = f(t, T)S^K$
Using Black Schole's PDE, the Portfolio will have PDE given by

$$r\pi = \frac{\partial \pi}{\partial t} + \frac{1}{2} \frac{\partial^2 \pi}{\partial S^2} \sigma^2 S_t^2 + r \frac{\partial \pi}{\partial S} S_t \text{-----(Eq 1)}$$

$$\frac{\partial \pi}{\partial t} = \frac{\partial f}{\partial t} S^K, \quad \frac{\partial \pi}{\partial S} = fK S^{K-1}, \quad \frac{\partial^2 \pi}{\partial S^2} = fK(K-1) S^{K-2}$$

Substituting the above expressions in equation 1 we have

$$rfS^K = \frac{\partial f}{\partial t} S^K + \frac{1}{2} fK(K-1)S^{K-2}\sigma^2 S_t^2 + rfKS^{K-1}S_t$$

$$= S^K \left(\frac{\partial f}{\partial t} + \frac{1}{2} fK(K-1)\sigma^2 + rfK \right)$$

$$\text{Or } rf = \frac{\partial f}{\partial t} + \frac{1}{2} fK(K-1)\sigma^2 + rfK$$

[4]

- ii) The derivative is worth S_T^K at time T. Hence the portfolio value at $T = S_T^K$
This means $f(T, T) = 1$

[2]

- iii) We can solve the differential equation

$$rf = \frac{\partial f}{\partial t} + \frac{1}{2}fK(K-1)\sigma^2 + rfK \quad (0.5)$$

The differential equation can be written as

$$r = \frac{\frac{\partial f}{\partial t}}{f} + \frac{1}{2}K(K-1)\sigma^2 + rK \text{ or } \frac{\frac{\partial f}{\partial t}}{f} = -\frac{1}{2}K(K-1)\sigma^2 - r(K-1)$$

The solution to this equation is $\ln(f) = \left(-r(K-1) - \frac{1}{2}\sigma^2K(K-1)\right)t + C$

Where C is a constant.

Now at $t=T$, $f=1$, which means $\ln(f)=0$

$$\text{This means } 0 = \left(-r(K-1) - \frac{1}{2}\sigma^2K(K-1)\right)T + C$$

$$\text{Or } C = \left(r(K-1) + \frac{1}{2}\sigma^2K(K-1)\right)T$$

$$\begin{aligned} \text{So that } \ln(f) &= \left(-r(K-1) - \frac{1}{2}\sigma^2K(K-1)\right)t + \left(r(K-1) + \frac{1}{2}\sigma^2K(K-1)\right)T \\ &= \left(-r(K-1) - \frac{1}{2}\sigma^2K(K-1)\right)(T-t) \end{aligned}$$

$$\text{Or } f = \exp\left(\left(-r(K-1) - \frac{1}{2}\sigma^2K(K-1)\right)(T-t)\right)$$

[4]

- iv) The price of the security at time t is $\pi(S, t) = f(t, T)S^K$
Using Ito calculus,

$$\frac{\partial \pi}{\partial S} = K\pi/S, \frac{\partial^2 \pi}{\partial S^2} = K(K-1)\pi/S^2, \frac{\partial \pi}{\partial t} = -\left[r(K-1) + \frac{\sigma^2K(K-1)}{2}\right]fS^K = -\left[r(K-1) + \frac{\sigma^2K(K-1)}{2}\right]\pi$$

$$\text{Now } d\pi = \frac{K\pi}{S}dS + \frac{\frac{1}{2}K(K-1)\pi}{S^2}(dS)^2 - \pi\left[r(K-1) + \frac{\sigma^2K(K-1)}{2}\right]dt \text{ where}$$

$$dS_t = S_t \sigma dW_t + S_t \mu dt, (dS)^2 = \sigma^2 S_t^2 dt,$$

$$d\pi = \pi(K\mu - r(K-1))dt + K\pi\sigma dW_t$$

For the process to be martingale, the drift term should be zero. i.e. $K\mu = r(K-1)$ or

$$\mu = r(K-1)/K$$

[4]

[14 Marks]

Solution 6:

Working is done in 1000's

Assume that $U(1000)=1$ and $U(0) = 0$

(Note: Any assumption is acceptable as this will affect only the scale on y-axis)

From statement (I), $U(700) = 0.25U(0) + 0.75 U(1000) = 0.75$

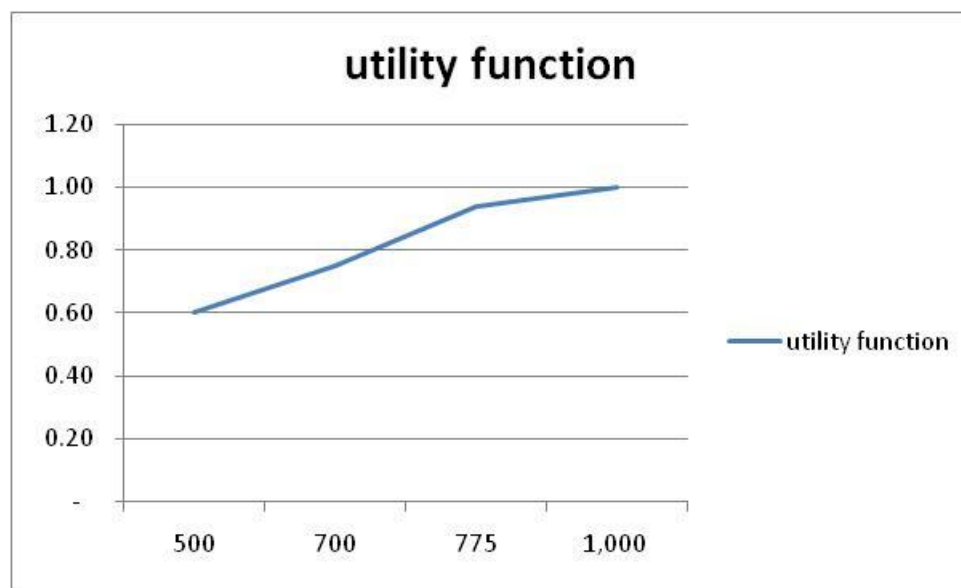
From statement (II), the person is willing to take a risk on 500. The expected wealth after allowing for insurance premium is $[350 \text{ (certain)} + 500 \times 0.85 \text{ (expected value of the risk taken)}]$

$U(775) = 0.15U(500) + 0.85 U(1000) \dots (i)$

From comment on the gamble, $U(500) = 0.4U(0) + 0.60 U(1000) = 0.60$

Substituting value of $U(500)$ in (i), we have $U(775) = 0.94$

Chart-



[5 Marks]

Solution 7:

- i) Using risk neutral valuation, the value of three month European stock option is $[p \cdot \text{value}(400) + (1-p) \cdot \text{value}(200)] / 1.02$
This has been worked out as follows:

Stock Price	250
Strike Price	250
Stock price if increases	400
Stock price if decreases	200
T	1 quarter
Interest per annum	8%
Quarterly Interest	2%
u	1.6
d	0.8
p	0.275

		Payoff
		0
Value		400
35.54	250	
		200
		50
		Payoff

[3]

- ii) The value of the option after 3 months period is either 0 (if the stock price is Rs. 400) or 50 (if the stock price is Rs. 200)
Let us consider a portfolio consisting of $-\Delta$ shares and 1 option.

The value of the portfolio is either
 $-200 \Delta + 50$
 Or -400Δ

This would mean $-200 \Delta + 50 = -400 \Delta$
 Or $\Delta = -0.25$

Which means the value of the portfolio is certain to be 100. Hence the risk free value of the portfolio is Rs. 100 ...(Step A)

The current value of the portfolio = $250 * 0.25 + P$ where P is the value of the option,

Since the portfolio must earn risk free rate of interest,

Value of portfolio = $(250 * 0.25 + P) * 1.02$

Using results of Step A and No-Arbitrage principle,

$(250 * 0.25 + P) * 1.02 = 100$

Or $(62.5 + P) * 1.02 = 100$

Or $63.75 + 1.02P = 100$ or $1.02P = 36.25$ or $P = 35.54$

[4]

[7 Marks]**Solution 8:**

- i) The effective fall in return of the portfolio in next 6 months = $-10\% + 2\% = -8\%$ i.e. -16% per annum.

This means $E(R_p) - r = -16\% - 6\% = -22\%$

From the CAPM equation, $E(R_p) = r + \beta E(R_M - r)$

Or $E(R_M - r) = [E(R_p) - r] / \beta = -22\% / 1.5 = -15\%$

This means $E(R_M) = -9\%$

Or market index return drops by -12% per annum so that the effective return considering 3% dividend yield = -9%

Now 12% drop in value per annum indicates a drop in value of 6% in next 6 month time. This means the Sensex will fall from current value of 25,000 to 23,500 in next 6 months.

Since the ratio = $50\text{cr}/25000 = 20,000$ it means for hedging the portfolio manager needs to buy 20,000 put options with an exercise price of 23,500

Stock price at time 0	25000
K	23500
T-t	0.5
Volatility	0.3
r	6.0%
q	0.03
d1	$[\ln(ST/K)+(r-q+\sigma^2)(T-t))/[\sigma*(T-t)^{0.5}]$
	0.468460146
d2	$d1-\sigma(T-t)^{0.5}$
	0.256328111
N(d1)	0.680272217
N(d2)	0.601151254
N(-d1)	0.319727783
N(-d2)	0.398848746
Value of one put option	$K\exp(-r(T-t))N(-d2) - S_0\exp(-q(T-t))N(-d1)$
	1221.74173

This means total cost of buying 20,000 options = Rs. 2.44 cr

[7]

ii) From the put call parity, we have

$$S_0e^{-qT} + p = c + Ke^{-rT}$$

Or a put option can be recreated from the following portfolio

$$c + Ke^{-rT} - S_0e^{-qT}$$

This means he has to invest in 20,000 call options with strike price 23,500, maturing in 6 months' time

Sell $50\text{cr} \times \exp(-0.03 \times 0.5) = 49.2$ cr worth of stock, buy 20,000 call options costing 6.09 cr and invest 45.61 cr in risk free rate

Invest remainder in risk free rate of 6.5% = 22,805

[3]

iii) Volatility of the market portfolio = $1.5 \times .30 = 0.45$

We need to buy risk-free securities in order to delta hedge the portfolio.

Delta of the put option = $\exp(-qt)[N(d1)-1]$

= -0.295

i.e. 29.5% of the portfolio i.e. 14.7 cr. should be sold and invested in risk-free securities.

Stock price at time 0	500,000,000
K	450,000,000
T-t	0.5
Volatility	0.45
r	6.0%
q	0.04
d1	$[\ln(ST/K)+(r-q+\sigma^2)(T-t)]/[\sigma*(T-t)^{0.5}]$
	0.52164215
d2	$d1-\sigma(T-t)^{0.5}=0.203444098$
N(d1)	0.699040245
N(d2)	0.580606032
N(-d1)	0.300959755
N(-d2)	0.419393968

[5]

[15 Marks]

Solution 9:

Property	Whether CIR Model satisfies this
The model should be arbitrage-free.	Yes
Interest rates should be positive.	Yes
Interest rates should be mean-reverting over the long term.	Yes
Bonds and derivative contracts should be easy to price.	Yes
The model should produce realistic interest rate dynamics.	No (does not model periods of high/low volatility)
It should fit historical interest rate data adequately.	No
It should be easy to calibrate to current market data.	No
It should be flexible enough to cope with a range of derivatives.	No (can be used to price simple options but not complex derivatives)

[6 Marks]

Solution 10:

- i) Under Merton Model, determine the probability of default on the debt raised by the company clearly stating the assumptions made.

Assumption:

The assets of the company follow geometric Brownian motion. Hence the value of the assets of the company can be described by Log- Normal Distribution.

If L_t is the loan value, F_t is the gross asset value, E_t is the net asset value at time t and the loan matures at time T

It follows that E_t is the value of a call option on the gross assets with strike L and

$$F_t = E_t + L_t$$

$$E_0 = \text{Rs. 100 crore}$$

$$\sigma = 25.00\%$$

$$r = 8.00\%$$

$$L = \text{Rs. 90 crore}$$

$$(T - t) = 5$$

Value of E_0 is given by Black-Scholes Option pricing formula i.e.

$$E_0 = F_0 N(d_1) - L e^{-rT} N(d_2)$$

$$d_1 = \left(\ln \left(\frac{F_0}{L} \right) + (r + 0.5\sigma^2)(T - t) \right) / (\sigma \sqrt{(T - t)})$$

$$d_2 = d_1 - \sigma \sqrt{(T - t)}$$

This has to be solved by trial values for F_0 and then interpolating the values to obtain correct F_0

Trial Value for F_0	(d_1)	(d_2)	$N(d_1)$	$N(d_2)$	Estimated E_0
130	1.6529	1.0938	0.9508	0.8630	71.54
190	2.3317	1.7727	0.9901	0.9619	130.10

Using linear interpolation

$$\frac{(E_0 - 71.54)}{(130.10 - 71.54)} = \frac{(F_0 - 130)}{(190 - 130)}$$

Solving for F_0 when E_0 is 100, gives $F_0 = 159.16$.

Note: The actual value for F_0 is 159.5. as this value depends on the initial guess value, any answer with correct steps and calculation is given equal credit.

The probability of default is same as the probability that the call option is not exercised.

For a call option, probability of exercising the option is $N(d_2)$. Hence probability that the option will not be exercised is $1 - N(d_2)$

For $F_0 = 159.16$, $1 - N(d_2) = 7.27\%$

For $F_0 = 159.50$, $1 - N(d_2) = 7.22\%$

[10]

ii) Determine the yield on the Zero coupon bonds.

Given the $F_0 = 159.50$ and $E_0 = 100.00$ we have $L_0 = 59.50$

Therefore annual yield is given by $\left(\frac{90}{59.5} \right)^{1/5} = 1.0863$ i.e. 8.63%

[1.5]

iii) Recovery Rate(δ)

The credit spread needs to be 25 basis points. Hence the yield on the ZCB is 8.25%

The equation of value for the bond is given by

$$59.5 = (7.22\% \times (\delta) \times 90 + (1 - 7.22\%) \times 90) \times (1.0825)^{-5}$$

Solving for (δ), we have (δ) = 76.01%

[2.5]

iv)

a. Structural Model:

- Structural models are explicit models for a corporate entity issuing both equity and debt. They link default events explicitly to the fortunes of the issuing corporate entity.
- The models are simple and cannot be realistically used to price credit risk. However, studying them does give an insight into the nature of default and the interaction between bondholders and shareholders.
- These models are called “structural” because they focus on the financial structure – the split between debt and equity – of the company issuing the bond.
- An example of a structural model is the Merton model.

b. Reduced Form Models:

- These models use observed market statistics related to the company like the credit ratings issued by credit rating agencies such as Standard and Poor’s and Moody’s.
- The credit rating agencies will have used detailed data specific to the issuing corporate entity when setting their rating.
- They will also regularly review the data to ensure that the rating remains appropriate and will re-rate the bond either up or down as necessary.
- Reduced-form models use market statistics along with data on the default-free market to model the movement of the credit rating of the bonds issued by a corporate entity over time.
- The output of such models is a distribution of the time to default.
- These models are called “reduced-form” because they ignore any information relating to the particular company issuing the bond. Instead, they model the different levels of creditworthiness and how companies move from one status to another.

c. Intensity Based Models:

- They typically model the “jumps” between different states (usually credit ratings) using transition intensities.
- This approach uses a continuous-time multiple state model (jump process) where the states correspond to the creditworthiness of the company.
- The “intensities” (denoted by λ) are the rates driving the company to switch from one state to another as time passes.
- Intensity-based models are an example of a reduced-form model.

[5]

[19 Marks]

Solution 11:

- i) For capital market line, risk is measured by σ . In a SML, risk is measured by β . As such CML is applicable only to an investor's final portfolio (which is efficiently diversified, with no Specific risk. SML is applicable to any security, asset or portfolio which may contain both components of risk.

[2]

- ii) In a CAPM world every asset lies on the SML.

By definition, for any portfolio p, $\sigma(r_p, r_m) = \rho(r_p, r_m) \sigma(r_p) \sigma(r_m)$

It follows that $\beta_{p,m} = \sigma(r_p, r_m) / \sigma^2(r_m) = \rho(r_p, r_m) \sigma(r_p) \sigma(r_m) / \sigma^2(r_m) = \rho(r_p, r_m) \sigma(r_p) / \sigma(r_m)$

It follows that SML can be written as $E(r_p) = r_f + (E(r_m) - r_f) \rho(r_p, r_m) \sigma(r_p) / \sigma(r_m)$

Comparing this equation with CML which holds for all efficient portfolios say F,

$$E(r_F) = r_f + (E(r_m) - r_f) \sigma(r_F) / \sigma(r_m)$$

It is clear that the portfolio lies on SML and CML if $\rho(r_p, r_m) = 1$ otherwise not i.e. p which lies on security market line (SML) is not an efficient combination of the risk-free asset and the market portfolio if $\rho[r_p, r_m] < 1$

[5]

[7 Marks]
